Butun sonlarni ko'paytirish, bo'lish va ular ustida amallar

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Qaysi ifodaning qiymati manfiy?**

A) $-(-2)^3$ **B)** $(-2)^4$

C) $(-2)^2 \cdot (-1)^2$ **D)** -2^4

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2. $-3^2 + (-3)^2$ **ni hisoblang.**

A) 9 **B)** 18 **C)** −18 **D)** 0

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3. $20 - 36 : (-4) \cdot 2$ **ni hisoblang.**

A) 2 **B)** 38 **C)** −38 **D)** 8

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4. $(-5) \cdot (-2)^3$ **ni hisoblang.**

A) 40 **B)** −40 **C)** 1000 **D)** −1000

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5. $[(-3)^2 - 15] : [2 - (-4)] \cdot (-3)$ **ni hisoblang.**

A) −1 **B)** −3 **C)** 3 **D)** 12

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6. $\sqrt[3]{-3375} : 5 + \sqrt{256} \cdot (-1)^3$ **ni hisoblang.**

A) 13 **B)** −13 **C)** −19 **D)** 19

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7. **Qaysi tenglik noto'g'ri?**

A) $-\sqrt{36} = -6$ **B)** $\sqrt{36} = -6$

C) $\sqrt{36} = 6$ **D)** $\pm\sqrt{36} = \pm 6$

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8. **Qaysi ifoda ma'noga ega emas?**

A) $-\sqrt{64}$ **B)** $\sqrt{-64}$ **C)** $0 : (-64)$ **D)** $\sqrt[3]{-64}$

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9. $(-36) : 4 + (-2) \cdot 3$ **ifodaga qavs qo'yib, 18 hosil qiling. Qaysi biri to'g'ri?**

A) $(-36) : [4 + (-2) \cdot 3]$ **B)** $(-36) : [4 + (-2)] \cdot 3$

C) $(-36) : \{[4 + (-2)] \cdot 3\}$ **D)** $[(-36) : 4 + (-2)] \cdot 3$

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10. $(-48) : \square \cdot (-2) = -32$ **bo'lsa, □ nechaga teng?**

A) 6 **B)** 3 **C)** −6 **D)** −3

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11. $(-1) \cdot (-2) \cdot (-3) \cdot (-4) \cdot (-5)$ **ni hisoblang.**

A) 120 **B)** −120 **C)** −15 **D)** 15

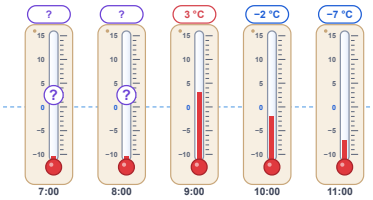
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12. **Samolyot uchib ketayotganda yerda harorat +9 °C edi. Samolyot ko'tarilgani sari tashqi harorat har 1 000 m da 6 °C ga pasayadi. 4 000 m balandlikda tashqi harorat necha gradus?**

A) −24 °C **B)** 33 °C **C)** −15 °C **D)** 15 °C

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13. **Rasmda harorat har soatda bir xil miqdorda o'zgargan. Soat 7:00 da harorat necha gradus bo'lgan?**



A) 18 °C **B)** −7 °C

C) 8 °C **D)** 13 °C

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14. **Dartsda nishon markaziga tekkan o'q uchun +5 ochko, nishonga umuman tegmagan o'q uchun −3 ochko, nishon chetiga tekkan o'q uchun 0 ochko. Ali 12 ta o'q otdi: 8 tasi markazga tegdi, 3 tasi nishonga tegmadi, 1 tasi chetiga tegdi. U necha ochko to'pladi?**

A) 40 **B)** 49 **C)** 31 **D)** 28

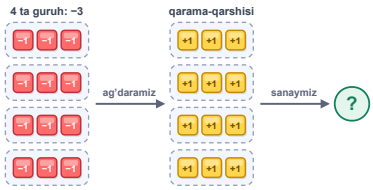
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15. **Robot o'yinchoq har bir to'g'ri buyruqda 4 qadam oldinga, har bir xato buyruqda 2 qadam orqaga yuradi. Malika unga 15 ta buyruq berdi, robot boshlang'ich joyidan 36 qadam oldinga ketdi. Nechta buyruq xato bo'lgan?**

A) 4 **B)** 3 **C)** 6 **D)** 11

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16. **Rasmdagi algebraik tugmalar qaysi ko'paytmani ko'rsatadi? (Sariq — +1, qizil — −1.)**

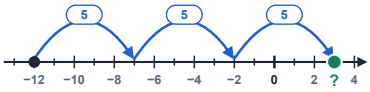


A) $(-4)\cdot(-3)=12$ **B)** $(-4)\cdot(-3)=-12$

C) $4\cdot(-3)=-12$ **D)** $(-4)+(-3)=-7$

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17. **Rasmdagi sakrashlarga qaysi ifoda mos?**



A) $-12+3\cdot5=3$ **B)** $-12-3\cdot5=-27$

C) $-12\cdot3+5=-31$ **D)** $(-12+3)\cdot5=-45$

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18. **Qaysi hisob to'g'ri?**

A) $12:(-2)\cdot3=-2$ **B)** $-4^2=16$

C) $\sqrt{25}=\pm5$ **D)** $(-3)^3=-27$

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19. $(-6)^2:(-4)-(-2)^3\cdot(-1)$ **ni hisoblang.**

A) 1 **B)** −17 **C)** 17 **D)** −1

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20. $a=-3$, $b=2$ **bo'lsa,** $a^2-b^3\cdot a$ **ning qiymatini toping.**

A) −33 **B)** −15 **C)** 15 **D)** 33

Javoblar

Butun sonlarni ko'paytirish, bo'lish va ular ustida amallar · 2-test (qiyinroq)

№	Javob	Qiymati
1	**D**	-2^4
2	**D**	0
3	**B**	38
4	**A**	40
5	**C**	3
6	**C**	−19
7	**B**	$\sqrt{36} = -6$
8	**B**	$\sqrt{-64}$
9	**A**	$(-36) : [4 + (-2) \cdot 3]$
10	**D**	−3

№	Javob	Qiymati
11	**B**	−120
12	**C**	−15 °C
13	**D**	13 °C
14	**C**	31
15	**A**	4
16	**A**	$(-4) \cdot (-3) = 12$
17	**A**	$-12 + 3 \cdot 5 = 3$
18	**D**	$(-3)^3 = -27$
19	**B**	−17
20	**D**	33

Qisqa yechimlar

1. $-2^4 = -16$ — qavs yo'q, minus darajaga ko'tarilmaydi. Javob: -2^4.
Qolganlari: $(-2)^4 = 16$, $4 \cdot 1 = 4$, $-(-8) = 8$.

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2. $-3^2 = -9$, $(-3)^2 = 9$.
$-9 + 9 = 0$. Javob: **0**. (18 — ikkalasini ham 9 deb olinsa.)

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3. Ko'paytirish va bo'lish chapdan o'ngga: $36 : (-4) = -9$, $(-9) \cdot 2 = -18$.
$20 - (-18) = 38$. Javob: **38**.
(2 — ishora xato, 8 — avval ayirilgan.)

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4. Avval daraja: $(-2)^3 = -8$.
$(-5) \cdot (-8) = 40$. Javob: **40**.
(−1 000 — avval ko'paytirib, keyin darajaga ko'tarilgan: $(-10)^3$.)

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5. Birinchi qavs: $(-3)^2 - 15 = 9 - 15 = -6$. Ikkinchi qavs: $2 - (-4) = 2 + 4 = 6$.
Chapdan o'ngga: $(-6) : 6 = -1$, $(-1) \cdot (-3) = 3$.
Javob: **3**. (12 — $(-3)^2$ o'rniga −9 olingan.)

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6. Ildiz va daraja: $\sqrt[3]{-3375} = -15$ (chunki $(-15)^3 = -3375$), $\sqrt{256} = 16$, $(-1)^3 = -1$.
$(-15) : 5 = -3$, $16 \cdot (-1) = -16$.
$-3 + (-16) = -19$. Javob: **−19**.

---

7. $\sqrt{\ }$ yozuvi faqat musbat ildizni bildiradi: $\sqrt{36} = 6$.
Noto'g'risi: $\sqrt{36} = -6$. −6 ni $-\sqrt{36}$ deb yozamiz.

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8. Har qanday sonning kvadrati manfiy emas, shuning uchun $\sqrt{-64}$ ma'noga ega emas.
Qolganlari: −4, −8, 0.

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9. Qavs ichida: $4 + (-6) = -2$, $(-36) : (-2) = 18$. Javob: $(-36) : [4 + (-2) \cdot 3]$.
Qolganlari: −54, −33, −6.

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10. Tekshiramiz: $(-48) : (-3) = 16$, $16 \cdot (-2) = -32$ ✓.
3 bo'lsa: $-16 \cdot (-2) = 32$ — mos emas. Javob: **−3**.

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11. Juft-juft: $(-1)\cdot(-2)=2$, $2\cdot(-3)=-6$, $(-6)\cdot(-4)=24$, $24\cdot(-5)=-120$.
Javob: **−120**. (−15 — yig'indi.)

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12. 4 000 : 1 000 = 4 marta pasayadi: $4\cdot(-6)=-24$.
9 + (−24) = **−15 °C**. (−24 °C — yerdagi +9 °C qo'shilmagan.)

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13. Har soatda o'zgarish: −2 − 3 = −5 °C.
7:00 — 9:00 dan 2 soat oldin: $(-2)\cdot(-5)=10$.
3 + 10 = **13 °C**. (8 °C — bu 8:00 dagi harorat.)

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14. $8\cdot5+3\cdot(-3)+1\cdot0=40-9+0=31$.
Javob: **31**. (28 — chetiga tekkan o'q ham −3 deb olingan.)

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15. Hammasi to'g'ri bo'lsa: $15\cdot4=60$ qadam. Har bir xato buyruq natijani $4+2=6$ qadamga kamaytiradi: $60-36=24$, $24:6=4$.
Xato: **4**. Tekshiramiz: $11\cdot4+4\cdot(-2)=44-8=36$.

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16. −3 dan (uchtadan qizil) 4 ta guruh olinib, ag'darilgan — bu $(-4)\cdot(-3)$.
12 ta sariq tugma: $(-4)\cdot(-3)=12$.

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17. Boshi −12, o'ngga 3 marta 5 dan sakrash: $3\cdot5=15$.
$-12+15=3$. Javob: $-12+3\cdot5=3$.

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18. $(-3)^3=-27$ — to'g'ri.
$-4^2=-16$; $12:(-2)\cdot3=(-6)\cdot3=-18$; $\sqrt{25}=5$ (faqat musbat).

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19. Darajalar: $(-6)^2=36$, $(-2)^3=-8$.
$36:(-4)=-9$, $(-8)\cdot(-1)=8$.
$-9-8=-17$. Javob: **−17**.

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20. $a^2=(-3)^2=9$, $b^3=8$, $8\cdot(-3)=-24$.
$9-(-24)=33$. Javob: **33**.
(−15 — ayirishda ishora unutilgan, 15 — -3^2 deb olingan.)
